STRING PYTHON CHALLENGES AND CODE

1.Length of a string

```
 a=input()
 print(len(a))
```

2.Join the strings

```
 a=input()
  b=input()
  print(a+" "+b)
```

3.Compare the Strings

```
a=input()
b=input()
if a==b:
  print("true")
else:
  print("false")
```

4.First character of a string

```
a=input()
print(a[0])
```

5.Last character of a string

```
a=input()
print(a[-1])
```

6.Join first & last characters of a string

```
a=input()
print(a[0]+a[-1])
```

7. Alphanumeric strings

```
a=input()
l=n=False
for i in a:
  if i.isalpha():
    l=True
  elif  i.isdigit():
    n=True
if l and n:
  print("true")
else:
  print("false")
```

8.Uppercase check

```
a=input()
if a.isupper():
  print("true")
else:
  print("false")
```

9.Lowercase check

```
a=input()
if a.islower():
  print("true")
else:
```

```
  print("false")
```
10.String lengths equal or not
```
a=input()
b=input()
c=len(a)
d=len(b)
if c==d:
  print("true")
else:
  print("false")
```
11.Reverse a string
```
a=input()
print(a[::-1])
```
12.Split strings by comma
```
a=input()
a.split(",")
print("\n".join(a.split(",")))
```
13.Remove vowels
```
a=input()
for i in a:
  if i not in "aeiouAEIOU":
    print(i,end="")
```
14.Reverse the character case
```
a=input()
print(a.swapcase())
```
15.Print ascii value
```
a=input()
print(ord(a))
```
16.Remove occurrences of a char
```
a=input()
occur=input()
for i in a:
  if i==occur:
    continue
  else:
    print(i,end="")
```
17.Search for a word
```
a=input()
b=input()
if b in a:
  print("true")
else:
  print("false")
```
18.Count vowels and consonants
```
a=input()
c=0
s=0
for i in a:
```

```
  if i in "aeiouAEIOU":
    c+=1
  else:
    s+=1
print(c)
print(s)
```

19.Anagram check

```
a=input()
b=input()
a=a.replace(" ","").upper()
b=b.replace(" ","").upper()
if sorted(a)==sorted(b):
  print("true")
else:
  print("false")
```

20.Palindrome check

```
a=input()
if a==a[::-1]:
  print("true")
else:
  print("false")
```

digit base qusestions

1.Count the number of digits

```
n=int(input())
if n==0:
  print(1)
else:
  count=0
  while n>0:
    count+=1
    n//=10
  print(count)
```

2.Sum of Even Digits

```
n=int(input())
sum1=0
while n>0:
  last=n%10
  if last%2==0:
    sum1+=last
  n//=10
print(sum1)
```

3.Product of Digits

```
n=int(input())
summ=1
while n>0:
  last=n%10
  summ*=last
```

```
  n//=10
print(summ)
```
4.Reverse a Number
```
n=int(input())
rev=0
while n>0:
  last=n%10
  rev=rev*10+last
  n//=10
print(rev)
```
5.Palindrome Number
```
n=int(input())
temp=n
rev=0
while temp>0:
  last=temp%10
  rev=rev*10+last
  temp//=10
if n==rev:
  print("Palindrome")
else:
  print("Not Palindrome")
```
6.First Digit of a Number
```
n=int(input())
rev=0
while n>0:
  r=n%10
  rev=rev*10+r
  n//=10
print(rev%10)
```
7.Count Zeros in a Number
```
n=int(input())
temp=n
summ=0
while n>0:
  r=n%10
  fact=1
  for i in range(1,r+1):
    fact=fact*i
  summ=summ+fact
  n//=10
if summ==temp:
  print("Strong")
else:
  print("Not Strong")
```
10.Check Perfect Number
```
n=int(input())
summ=0
```

```
for i in range(1,n):
  if n%i==0:
    summ+=i
if summ==n:
  print("Perfect")
else:
  print("Not Perfect")
```

Operators:

1.Input and Output Practice

```
a=int(input())
b=float(input())
c=input().strip()
d=int(input())
e=float(input())
print(a)
print(b)
print(c)
print(d)
print(e)
```

2.Progressive Float Execution

```
import struct
a=float(input())
a=struct.unpack('f',struct.pack('f',a))[0]
print(f"{a:.5f}")
print(f"{a:.4f}")
print(f"{a:.3f}")
print(f"{a:.2f}")
print(f"{a:.1f}")
print(f"{a:.0f}")
```

3.Problem: Sum, Difference, Product, Division and Modulo.

```
a,b=map(int,input().split())
print(a+b)
print(a-b)
print(a*b)
print(int(a/b))
if a<0 or b<0:
  c=abs(a)%abs(b)
  print(-1*c)
else:
  print(a%b)
```

4.ASCII

```
a=int(input())
print(chr(a))
```

5.Update Counter

```
a=int(input())
b=a+5
c=b*2
d=c-3
```

```
print(d)
```
6.ODD or EVEN
```
num=int(input())
if num%2==0:
  print("Even")
else:
  print("Odd")
```
7.Greatest Among Three Numbers
```
a=int(input())
b=int(input())
c=int(input())
if a>b and a>c:
  print(a)
elif b>a and b>c:
  print(b)
else:
  print(c)
```
8.Character Digit to Integer
```
a=input()
print(int(a))
```
9.Personalized Greeting
```
n=input()
print(f"Hi {n}, welcome to the programming class!")
```
10.Calculate Total and Average Marks
```
name=input()
a=int(input())
b=int(input())
c=int(input())
d=int(input())
e=int(input())
total=a+b+c+d+e
print("Student Name:",name)
print("Total Marks:",total)
print("Average Marks:",f"{total/5:.2f}")
```
11.Shape Area Calculation
```
a=int(input())
b=int(input())
c=int(input())
r=int(input())
e=int(input())
h=int(input())
total=3.14*r*r
print("Area of Square=",a*a,sep="")
print("Area of Rectangle=",b*c,sep="")
print("Area of Circle=",f"{total:.2f}",sep="")
print("Area of Triangle=",f"{(1/2)*e*h:.0f}",sep="")
```
12.Shape Perimeter Calculation Program
```
a=int(input())
```

```
l=int(input())
w=int(input())
r=int(input())
base1=int(input())
side=int(input())
base2=int(input())
print(4*a)
print(2*(l+w))
print(f"{2*3.14*r:.2f}")
print(base1+side+base2)
```

13.Handling Mixed Inputs

```
a=int(input())
b=input()
print(a)
print(b,sep="")
```

Patterns:

1.Print the pattern given below, using loops.

```
*****
*****
*****
*****
```

```
for i in range(1,5):
  print("*"*5)
```

2.Print the pattern given below, using loops

```
*
**
***
****
*****
```

```
for i in range(1,6):
  print("*"*i)
```

3.Print the pattern given below, using loops

```
1
12
123
1234
12345
```

```
for i in range(1,6):
  for j in range(1,i+1):
    print(j,end="")
  print("")
```

4.Print the pattern given below, using loops

```
1
22
333
4444
```

55555

```
for i in range(1,6):
  for j in range(1,i+1):
    print(i,end="")
  print("")
```

5.Print the pattern given below, using loops

1
2 3
4 5 6
7 8 9 10
11 12 13 14 15

```
n = 1
for i in range(1, 6):
    for j in range(i):
        print(n, end=" ")
        n += 1
    print()
```

6.Print the pattern given below, using loops

1
01
101
0101
10101
010101

```
for i in range(1, 7):
    for j in range(i):
        print((i + j + 1) % 2, end="")
    print()
```

7.Print the pattern given below, using loops

1
23
123
1231
23123

```
for i in range(1, 6):
    for j in range(i):
        print((j + 1) if i < 3 else ((j + 1) % 3 + 1), end="")
    print()
```

8.Print the pattern given below, using loops

5
5 4
5 4 3
5 4 3 2
5 4 3 2 1

```
for i in range(5, 0, -1):
    for j in range(5, i-1, -1):
        print(j, end=" ")
    print()
```

9.Print the pattern given below, using loops

```
5
4 5
3 4 5
2 3 4 5
1 2 3 4 5
```

```
for i in range(5, 0, -1):
    for j in range(i, 6):
        print(j, end=" ")
    print()
```

10.Print the pattern given below, using loops

```
    1
   12
  123
 1234
12345
```

```
for i in range(1, 6):
    print(" " * (5-i), end="")
    for j in range(1, i+1):
        print(j, end="")
    print()
```

11.Print the pattern given below, using loops

```
    *
   ***
  *****
 *******
*********
```

```
for i in range(1, 6):
    print(" " * (5-i) + "*" * (2*i-1))
```

12.Print the pattern given below, using loops

```
    2
   24
  246
 2468
246810
```

```
for i in range(1, 6):
    for j in range(1, i+1):
        print(j*2, end="")
    print()
```

13.Print the pattern given below, using loops

```
* * * *
*     *
*     *
* * * *
```

```
for i in range(4):
    for j in range(4):
        if i == 0 or i == 3 or j == 0 or j == 3:
            print("*", end=" ")
```

```
        else:
            print(" ", end=" ")
    print()
```

14.Print the pattern given below, using loops

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

```
for i in range(1, 6):
    print(" " * (5-i) + "*" * (2*i-1))

for i in range(4, 0, -1):
    print(" " * (5-i) + "*" * (2*i-1))
```